

Akus Chhabra

Brampton, ON | (416)-844-2587 | akuschh@gmail.com | [Portfolio](#) | [GitHub](#) | [LinkedIn](#) | Canadian Citizen

Education

Aerospace Engineering (B. Eng.)

Toronto Metropolitan University, Toronto, ON

September 2019 – April 2024

CGPA: 4.01/4.33 (with Distinction)

Skills

Programming Languages: Python, MATLAB (and Simulink), JavaScript, C, SQL, GraphQL, TypeScript

Software: Microsoft Office, Linux, Git, CATIA V5, Solidworks, MultiSim, Arduino, SpyderIDE, MongoDB, ANSYS Fluent, ANSYS Mechanical APDL, PyTorch, TensorFlow, Keras, Batch Script, Node.js, Shopify, Retool, React, HubSpot, Vercel, HTML, CSS, Postman, Shell Script, AWS S3, FFmpeg, Jira, FreshDesk, Java

Languages: English (Native), French (Fluent), Hindi, Punjabi

Professional Experience

Flight Operations Engineering & Support

May 2025 – Present

Skywise (Airbus), Waterloo, ON, Canada

- Design engine failure procedures that abide to regulations by certifying authorities including Transport Canada, EASA, ICAO, and FAA, ensuring safe flight operations for passengers
- Optimize flight paths through EFP analysis to ensure compliance with regulations while enabling customers to maximize revenue without penalizing the aircraft payload
- Develop software applications that automate manual data collection and obstacle analysis processes, significantly improving workflow efficiency and reducing processing time
- Optimize delivery timelines and backlog efficiency by managing and tracking updates in Jira, driving consistent KPI target progression
- Assess obstacle limitations to validate safe and efficient engine-out performance for customer operations

Full-Stack Developer

January 2025 – July 2025

Clouver, Toronto, ON, Canada

- Supported client operations on Retool using Shopify and JavaScript to streamline operations following DevOps principles and provide continuous enhancement of features based on client feedback
- Constructed queries using GraphQL to retrieve customer and order data to support client feature requests
- Developed order querying functionality with database platforms such as MongoDB to establish data versioning of existing orders, enabling clients to revert orders and measure business performance
- Integrated APIs such as Shopify and Google to develop app features such as address autocomplete, order file storage, and metadata queries
- Implemented a JS webhook from a SQL database to automate live video capture via FFmpeg, storing recordings in AWS S3 for client verification, with cross-platform support for Mac OS and Windows
- Synchronized data between the web applications Flex5 and HubSpot using REST API to maintain continuous client database identification

Aircraft Performance Engineer (Internship)

May 2022 – August 2023

Bombardier Aerospace, Toronto, ON, Canada

- Contrasted performance datasets using the pandas and matplotlib libraries in Python to validate changes made to the datasets and maintain continuity for future iterations
- Developed and updated AFM and FPCCM performance data for the Global aircraft using MATLAB to support pilots with their preliminary calculations and ensure aircraft operate within performance margins
- Managed simulated flight testing with suppliers to satisfy certification requirements for various aircraft conditions such as OEI, depressurization, and ETOPS
- Conducted mission profile analysis for the Global 7500 and 8000 aircraft to validate simulated flight test results delivered by suppliers

- Performed preliminary data analysis for the development of the Runway Overrun Awareness and Alerting System at various runway conditions, bleed, and braking conditions to prevent the risk of runway overruns
- Implemented git tracking using Linux to maintain version control of new and updated files

Projects

AOC Obstacle Parser App

June 2026 – July 2026

Waterloo, ON, Canada

- Engineered a Flask-based application with HTML and JavaScript automating aerodrome obstacle data collection and visualization, reducing manual processing time and improving KPIs
- Utilized GitHub for version control and documentation management, tracking changes, and supporting future software development.
- Designed and implemented a user-friendly interface that simplified application workflows, improving usability and reducing onboarding time for new users
- Optimized obstacle parsing process, decreasing AOC parsing time for datasets that previously required up to two hours to parse

Geodetic Transformation App

April 2026 – April 2026

Waterloo, ON, Canada

- Designed a Tkinter-based interface in Python to convert MGA94 Australian grid coordinates into WGS84 geographic coordinates, streamlining obstacle data processing
- Engineered a unit testing framework to ensure each function returns the expected result, reducing bugs and minimizing unintended behavior
- Validated EPSG calculations using mathematical models by Charles Karney, applying an extended Transverse Mercator projection to accurately compute geographic coordinates

PDF-Merger GUI

April 2025 – April 2025

Brampton, ON, Canada

- Developed a Python Tkinter-based GUI enabling users to merge PDF files in a custom order
- Packaged the application with cross-platform dependencies, reducing setup complexity for end-users
- Incorporated error handling to ensure supported file types are processed and provide user-friendly feedback
- Documented installation, usage, and troubleshooting steps to support maintainability and user onboarding
- Utilized GitHub to track development progress and maintain codebase integrity

Computer Vision Machine Learning Project

November 2023 – December 2023

Toronto Metropolitan University, Toronto, ON, Canada

- Trained a machine learning model in Python using Ultralytics YOLOv8 computer vision model to identify key components of various PCBs using PCB data images as the training source
- Classified image data into training, validation and testing data through semi-supervised learning to ensure performance metrics obtained from testing are not biased and prevent overfitting of data
- Illustrated performance metrics using a normalized confusion matrix, precision-confidence index and precision-recall index to analyze the model's predictive accuracy and enhance it through hyperparameter variation

Drone Payload Design Project

September 2023 – December 2023

Toronto Metropolitan University, Toronto, ON, Canada

- Prototyped a payload through the systems engineering development lifecycle to detect and disable a target drone all within a project budget of \$50
- Defined stakeholder requirements, constraints, and functions for the product design in addition to the methodology for the requirements loop and design loop to produce an effective solution
- Spearheaded the system prototyping team in developing the circuit schematics and implementing the physical components with the machine learning model
- Interfaced the ground unit to the payload using transmitters integrated by Arduino microcontrollers, successfully establishing the disabling mechanism for the prototype design